Equity and Resilience

A Semi-Systemic Literature Review

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1. Introduction

Equity and resilience have become common buzzwords thrown around with the advent of big data and artificial intelligence (AI) entering the workplace and our daily lives. The level of integration of this technology has created concerns about the equitable use of data and the resilience of AI models. Resilience is defined as, “the capacity to endure, adapt, and thrive amidst challenges and adversities” (Rane et al., 2024). This also includes topics particularly relevant to AI, such as sustainability, stability, and continuity. Equity is defined as, “the fairness and justice in policy, practice, and opportunity designed to address the distinct challenges of nondominant social groups” (Hendricks-Sturrup et al., 2023). For this document, equity will be used in the context of healthcare availability, Diversity, Equity, and Inclusion (DEI), organizational and criminal justice, and the argument of equity vs. equality. The purpose of this literature review is to analyze how AI and big data are being used, the equity and resilience concerns of current usage, and potential remedies and solutions to some of the issues. This review will also explore the connections between equity and resilience in unmanned aerial vehicles (UAVs) and how this emerging field currently analyzes equity in service and resilience in swarm drones. Since AI is such a novel technology, it’s essential to understand the current state of AI research, its inner workings, and its societal impacts (Patón-Romero et al., 2022).

1.1 Methods

A Semi-Systemic review made the most sense to go about the research due to the recency of the topic and the concerns of having multiple perspectives (Cachat-Rosset & Klarsfeld, 2023). To conduct the literature review, Google Scholar was used to find relevant documentation. Keywords such as “AI”, “Equity”, “Resilience”, “Race”, “Sex”, and “Justice” were used to find documents to review. For the UAV section, search terms included “UAV”, “Resilience”, “Equity”, and “Social Equity”. Documents were chosen by what was provided in class previously, their recency (all documents are from within the past ten years, with most of them coming from the past five years), and their content is relevant to topics surrounding equity and resilience with big data and AI. A limitation of this literature review is due to the nature of the AI problem being so new, there aren’t any historical sources to go off, and many of the papers are from the authors’ perspectives and may not be entirely objective.

2. Key Themes and Topics

To break down the topic, certain themes were derived that were covered across the papers collected. The themes that will be explored include the current uses of AI, how AI and resilience are connected, How AI and DEI are connected, ethics and societal concerns regarding AI, data considerations when using AI, and future considerations, particularly the human role in AI.

2.1 Current Uses of AI

AI has infiltrated many aspects of our everyday lives with applications such as ChatGPT and other AI assistants, such as Apple’s Siri, Microsoft’s CoPilot, and Google’s Gemini. While these are primarily large language models (LLMs) that aim to learn from the information fed to them and regurgitate the most logical results, much more sophisticated AI is being used in fields such as medicine, supply chain, criminal justice, and education. These topics will be explored in greater detail.

2.1.1 Healthcare and Medicine

AI is revolutionizing the field of healthcare by improving diagnostics, which allows doctors to analyze medical data in the past and identify diseases faster within the early stages (Rane et al., 2024). This allows doctors to begin interventions quicker, which could help save lives. The same paper discusses how machine learning (ML) algorithms are also being used in epidemiology, helping to predict disease outbreaks and maintain public health measures. Applications for AI also go into the operations of hospitals, focusing on how to optimize patient flow and care and advance telemedicine. Hospital efficiency and telemedicine saw a surge in importance during the Covid-19 pandemic that caused hospitals to fill up across the world and prevented doctors from reaching patients due to the safety lockdown restrictions. During the pandemic, AI was also used to track the virus's spread, analyze patient data, and help accelerate the development of a vaccine. AI currently plays a critical role in the healthcare supply chain by predicting demand, optimizing on-hand inventory levels, and identifying bottlenecks in the care processes. AI algorithms can analyze historical data to help predict when shortages of supplies will show up and suggest methods for working around them. On the side of telemedicine, AI is also used to increase the availability of mental health resources, providing personalized support and stress management tools (Rane et al., 2024).

AI can also be used in the prevention of diseases, not only from modeling historical data, but to vaccine and drug discovery by rapidly identifying new peptides, genes, and potential drug candidates for experimentation (Dankwa-Mullan et al., 2021). AI is even used to conduct surgeries, helping to reduce the effect of cognitive shortcuts and implicit biases (Johnson-Mann et al., 2021). With the use of AI, however, healthcare practitioners must be careful of biases present in current healthcare algorithms to ensure equitable healthcare (Ross, 2022). If unregulated, biases that guide population medicine, federal healthcare reimbursements such as Medicare and Medicaid, and clinical management can continue to discriminate against disenfranchised social groups (Thomasian et al., 2021). To ensure that healthcare AI incorporates standards of health equity, federal authorities are being urged to implement a regulatory framework that will protect the people (Gurevich et al., 2023). Currently, the FDA has approved AI applications under the "Software as Medical Devices" classification; however, many AI applications in healthcare are exempt from FDA review, despite already being in widespread use (Thomasian et al., 2021).

2.1.2 Supply Chain and Logistics

While mentioned briefly in the previous section, the applications of AI in the sector of supply chain and logistics have enabled companies to predict and manage disruptions, make better predictions of demand and supply, and allow for timely deliveries of materials and resources (Rane et al., 2024). Rane et al. also mention that supply chain analytics driven by AI can help anticipate seasonal trends, market conditions, and ever-chaining consumer behaviors, which allows companies to adjust inventory levels, schedules, and logistical plans. These adjustments can save the company money, time, and resources. In conjunction with the Internet of Things (IoT) provides up-to-the-minute information using real-time monitoring and AI-powered sensors. AI can coordinate and control vehicles and other machinery can make the entire logistics process nearly fully autonomous with little human intervention. AI also helps to assess supply chain risks and bottlenecks and can provide solutions to mitigate disruptions. In future applications, technologies such as blockchain can also be integrated with AI to provide greater transparency to help protect consumers (Rane et al., 2024).

2.1.3 Criminal Justice

In the criminal justice system, AI tools are being implemented in policing, bail decisions, and predicting recidivism rates (Lim, 2022). Location-based data is used to aggregate real-time information about crime frequency in different areas which allows police forces to determine how many police they should have on patrol in certain areas (Huq, 2019). Predictive algorithms are used to analyze massive amounts of data, and with ML technology, generate predictions on where and when criminal activity will occur. The biggest concern with AI and criminal justice is the amplification of biases in the justice system that already disenfranchise vulnerable groups (Abbas, 2022). AI and ML tools are trained from the data that they're fed, which is flawed to discriminate by race, sex, and socioeconomic status, which furthers oppression. An argument can be made for a multi-stratal approach that would help to prevent the further disenfranchisement of racially discriminatory practices. Another concern with the use of AL and ML for crime prevention is the issue of surveillance. The capabilities for around-the-clock surveillance monitored by an algorithm powered by biased data proposes ethical concerns, as well as concerns for overcontrolling sovereignty and the destruction of privacy in the name of law and order (Abbas, 2022).

2.1.4 Education

Conversely to AI in the criminal justice sector, AI is being used as a tool for the empowerment of equity and inclusivity in education. AI can recognize patterns, allowing LLMs to personalize learning by analyzing a student's learning styles and habits, their strengths and weaknesses in past work, and can create custom educational content to support their learning (Roshanaei et al., 2023). This allows for greater access to educational resources, as well as ensuring that students, regardless of their starting point, have an equal opportunity to learn and progress. The identification of learning gaps can also be handled by intelligent tutoring systems. By using data-driven decision-making processes, students can help identify gaps in their understanding.

Another use case for AI is increasing the accessibility of information. For example, differently-abled students can use AI speech-recognition technology to help the visually impaired learn (Roshanaei et al., 2023). Language learning and the spread of information across language barriers can also be enhanced with AI. AI-driven platforms can help to create global connections and collaboration in adaptive learning environments that otherwise wouldn't be possible. With these new connections, however, it's important to be aware of the potential barriers to equity that exist with the spread of information between different nations and cultures (Roshanaei et al., 2023).

2.2 Resilience and AI

Resilience, as defined previously as the ability to withstand, recover, and adapt to adverse situations, can be enhanced with AI by providing tools and strategies to anticipate, endure, adapt to, and overcome challenges and adversities (Rane et al., 2024). Four basic concepts of resilience include rebounding from trauma and returning to an equilibrium state, robustness, displaying grace under unexpected challenges that test a system's limits, and the ability to adapt to future surprises as conditions shift (Woods, 2015). These four concepts will be referenced throughout this section and are relevant to how AI can increase the resilience of other systems.

2.2.1 Resilience of Physical and Environmental Systems

AI has created tools for the maintenance and management of infrastructure that help to increase their resilience to disasters and other threats (Rane et al., 2024). This paper explains that when a building is being constructed, a Building Information Model (BIM) is created that provides information about the building. When coupled with AI and Augmented Reality (AR), construction progress can improve by increasing the clarity and immersion of BIMs. Like BIMs, digital twins are virtual models of buildings and other infrastructure that use real-time data and AI to help monitor infrastructure conditions and search for potential issues before they occur. The most common use for AI in infrastructure is for predictive analysis to try to anticipate failures and to allow for preventative maintenance (Rane et al., 2024). AI algorithms can parse large datasets, such as weather data and sensor inputs to predict the likelihood of extreme weather events and how they would impact a building. Taking a step back in infrastructure, AI is used for urban resilience through similar natural disaster predictions. Urban data can be crunched by AI to optimize infrastructure development and management on a larger scale than just a single building, which also has been expanded to transportation and energy (Rane et al., 2024). During health emergencies, such as a pandemic, large-scale data on infections can help improve our understanding of epidemiology and promote public health and safety interventions.

With less-than-satisfactory emission regulations globally, climate change has been a worsening issue that humanity will have to handle. The changes in overall climate, matched with more extreme and more frequent weather phenomena and natural disasters will soon be an aspect of life engineers, scientists, and policymakers will have to deal with. AI can also be applied to this sector in similar ways to previously mentioned; mostly through the prediction of future weather and climate trends and evaluating reduction and mitigation measures (Rane et al., 2024). AI can also optimize energy usage, reducing greenhouse gases and supporting more sustainable energy practices for industrial and commercial applications. Similarly to the use cases in infrastructure, digital twin models can be extended to simulate and analyze the effects of certain climate scenarios on buildings, transportation infrastructure, and entire cities, helping scientists and policymakers test resilient strategies for minimizing damage. AI can assist in supporting conservational efforts, especially with wildlife and ecological models (Rane et al., 2024).

2.2.2 Resilience for Societal and Community Wellness

Stemming from the previous topics of natural disaster and climate change resilience, disaster response can also be improved with help from AI. AI can analyze satellite imagery and social media far more efficiently than humans can by themselves, which helps to provide real-time insights into disaster events (Rane et al., 2024). This, in turn, can lead to faster recovery times, more effective allocation of resources, and faster assessment of impacts. Predictive modeling enhanced with AI, as mentioned previously, can also help increase the resilience of contingency plans. Disaster response teams can also utilize AI in search and rescue training, allowing for more holistic training and learning. Further integrating with the IoT can provide real-time data on environmental conditions and infrastructure status, allowing for better situational awareness for civilians as well as response members.

After a rescue mission, healthcare becomes the next essential step. AI in the healthcare domain can help strengthen the system’s capacity to handle adversity and react to influxes. This includes some of the previously mentioned tools for epidemiological research, helping to predict outbreaks and test the effectiveness of safety protocols. Diseases can be caught faster and earlier, the medical supply chain can react quicker to changes, and optimizations in hospital layout and operations can streamline the healthcare process, leading to quicker times between inpatient and outpatient care (Rane et al., 2024). For mental health resources, AI can provide chatbots and virtual therapists that provide immediate accessible support to those in need, further increasing the resilience of the healthcare system. This psychological resilience also includes the ability of AI-powered apps to personalize stress management and mindfulness exercises (Rane et al., 2024). On the management side, AI in cybersecurity can help to protect patients’ records and data, ensuring the continuity of care and services (Rane et al., 2024).

2.2.3 Resilience of Organizational and Economic Systems

AI tools are being used for business risk assessment, crisis management, and decision support, which allows for more resilient business practices. This mainly focuses on how predictive analytics can identify vulnerabilities in a company’s operations, oversee supply chain operations from multi-sensor data and market predictions, and enhance communications in crises (Rane et al., 2024). Many of these topics have been covered previously within other sectors; the same principles apply to business operations and supply chain logistics as the healthcare system does, for example. Another major use case for AI in the business sector is through cybersecurity enhancements. Machine learning algorithms can analyze network traffic for a business and detect anomalous behavior in real time, allowing for more secure networks (Rane et al., 2024). AI also currently supports the development of novel system protocols that are more secure, and AI systems can now automate threat responses, reducing the impact of cyberattacks on business and productivity (Rane et al., 2024).

2.3 Equity and AI

AI systems are becoming increasingly prevalent in many domains, as previously explored. These AI systems are more regularly making decisions that impact the lives of thousands of people (Cachat-Rosset & Klarsfeld, 2023). Ensuring that these systems operate equitably is essential for social justice. However, since humans hold biases that cross over into the training of AI through the data they feed, AI has begun to reflect societal inequalities (Renski et al., 2020). The same paper gives the example that algorithms trained on biased job referral data have resulted in fewer racial minorities being hired. Similarly, AI used in surgical settings can reinforce gender biases when they're trained on the preexistent data with predominantly male surgeons (Frigerio & Rashidian, 2023). When looking at some of the root causes of the biased data that AI models are trained on, the field of AI research is not very diverse. Cachat-Rosset and Klarsfeld found that across "…120 identified authors for the 46 selected guidelines, we found that 60.8% are male, 74.2% are white and 78.3% are nationals of Western countries (from North America, Europe, or Australia)." A more diverse representation in the AI development sector could lead to more equitable AIs that are better equipped to address DEI challenges.

While ethical guidelines are being developed to promote DEI with principles such as fairness, justice, and non-discrimination, the implementation of these principles is loose far from addressing the root of the issues (Cachat-Rosset & Klarsfeld, 2023). The concept of "equity by design" is brought up across some of the papers are a framework for building fair and inclusive AI systems (Patón-Romero et al., 2022). This includes carefully analyzing training data, algorithm design, and the implementation of AI to avoid biases. It's important to note the difference between equity and equality; while equality aims at providing the same resources for everyone, equity recognizes differing circumstances and allocates resources to those who need them more for all groups to each similar outcomes (Lim, 2022). By prioritizing equity, AI development can help uplift marginalized groups.

2.3.1 Gender Equity and AI

The intersection between gender equity and AI is a point in which there exist some current shortcomings, but also potential for advancing progress (Badarevski, 2023). The term "Gender by AI" is a broad term used to refer to how AI can be used as a tool to further gender equality, specifically as a tool to detect and address gender discrimination (Patón-Romero et al., 2022). Gender by AI also refers to how AI itself can perpetrate negative impacts on gender equality. In practice, this looks like biased AI systems, the underrepresentation of women in AI development (as tangentially approached in the last section), and ingrained gender stereotypes. This limited number of women in STEM has a direct negative impact on their impact on the development of AI that adequately addresses gender-specific needs and biases (Badarevski, 2023). Research has shown that there is even a gender gap in the adoption of generative AI; top female students were seen as more likely to opt to not use generative AI for assistance with classwork (Carvajal et al., 2024). The adoption and understanding of AI technology is becoming a valuable skill, and if top female students are avoiding its usage due to inherent biases in the algorithms, then AI will further limit the opportunities women can receive in the modern job market.

Some stereotypes and prejudices regarding women in STEM can further limit their opportunities within the AI sector. Since the data that AI is trained on is often biased, those biases are translated into unfair hiring practices that are handled by AI, and even AI virtual assistants have built-in stereotypes (Renski et al., 2020). Some initiatives have been proposed to increase gender inclusion in AI development, which is expected to bring in diverse perspectives, reduce cognitive biases in the design of algorithms, and help reduce bias-related consequences (Patón-Romero et al., 2022). In future work in AI, education and policies play a vital role in the preparation of the future generation to deal with gender biases. Badarevski suggests developing gender-equal educational programs and increasing the support of women's participation in STEM. An AI framework for analyzing and recognizing gender-biased data is also proposed, which would allow for the development of more inclusive and gender-sensitive interfaces.

2.3.2 Racial and Socioeconomic Equity and AI

AI's impact on racial and socioeconomic is primarily concerned with fairness and justice. As with gender disparities, AI systems can also heighten systemic racial injustice by perpetuating the preexistent biases within historical data (Dankwa-Mullan et al., 2021). An example of this is algorithms that use historical criminal records can disproportionately affect disenfranchised racial groups (Abbas, 2022). A similar, commonly used algorithm by police, healthcare workers, and insurance companies is a risk assessment algorithm. These types of algorithms are known to amplify racial bias if they're not carefully designed for equity (Thomasian et al., 2021). Another example of where this applies is in biased lending. In the 1960s, redlining was a practice in which lenders would deny lenders based on race, gender, nationality, and other factors (Cavazos, 2025). Since then, legislation has been passed making the practice illegal; however, since AI models are trained on historical data, they're using the biased lending data to make actual biased lending decisions in the modern day, which is being called "digital redlining" (Cavazos, 2025). AI fairness mostly compensates for how social group categories, such as ethnic groups, are socially constructed to avoid taking shortcuts that could be harmful (Cachat-Rosset & Klarsfeld, 2023).

Lim speaks about AI and how it affects intellectual property (IP). There is a lack of comprehensive methods for assessing how AI perpetuates inequality when it comes to IP laws. In this scenario, AI is being used to amplify historical inequalities due to a lack of social justice principles being implemented (Lim, 2022). Along similar lines of IP, since many AI models are proprietary technology, they have legal protections that end up slowing or in some cases entirely halting the auditing process to identify biases within their systems that affect marginalized racial and socioeconomic groups (Lim, 2022). Transparency and access to the underlying code of these algorithms are essential for finding and rectifying biased data and analysis.

When focusing more on socioeconomic factors, the term "digital divide" is used to describe the inequalities in access to technology brought about by socioeconomic disparities that affect students' access to an equal education (Roshanaei et al., 2023). The digital divide also includes students who cannot access AI technology. Socioeconomic and racial groups also receive different levels of diagnosis, treatment, and medical outcomes due to biases in healthcare-based AI systems (Gurevich et al., 2023). It's important to consider ethno-racial and socioeconomic backgrounds and understand the biases that exist within the data when developing AI for healthcare.

2.4 Ethical and Societal Concerns

Ethical and societal concerns with AI include algorithmic bias from flawed or misrepresentative data, which leads to discrimination and the sustained perpetuation of social inequalities across race, gender, and socioeconomic status (Cachat-Rosset & Klarsfeld, 2023; Preece et al., 2023). Critical issues of data privacy and security breaches loom over the industry that uses immense amounts of personal data to train its models (Akhtar et al., 2024). The "black box" problem, describing the lack of transparency and explainability in many AI models, further raises concerns about ethical practices and methods of holding AI developers and corporations accountable when AI systems make consequential decisions (Akhtar et al., 2024; Lin et al., 2021). This can have ramifications for human agency and autonomy, which severely impacts how the public trusts these technologies and the companies behind them (Akhtar et al., 2024; Hendricks-Sturrup et al., 2023). The adoption of more advanced AI models will bring forth more concerns with the ethical usage of AI and sensitive information, as well as create societal shifts that will affect the future of automation and work (Renski et al., 2020). Humanity will require careful consideration when it comes to the regulation, oversight, and maintenance of AI models to uphold humanistic values and principles (Dankwa-Mullan et al., 2021; Ross, 2022).

2.4.1 Data Considerations for AI

A data consideration that has been brought up across multiple articles is bias within training data. The semantics that are derived from the sources AI has access to are filled with human biases (Renski et al., 2020). AI and ML pattern recognition unintentionally latch onto the human biases that exist in historical data based on race, gender, and socioeconomic status (Ross, 2022). This bias can stem from the underrepresentation of certain populations in training data. For example, AI speech recognition tools make nearly twice as many errors in transcribing black speakers compared to white speakers due to this underrepresentation (Ross, 2022). Another example is how facial recognition software that is predominately trained on white people has greater difficulty in recognizing non-white faces (Avadhuta, 2020). It's essential to consider ethnoracial, sex, and gender characteristics when training AI because certain groups are more susceptible to maladies such as diseases that would otherwise go unnoticed (Gurevich et al., 2023). As the paper states, imbalanced data can lead to misclassifications, such as patients preferring Spanish being misclassified as speaking English.

The data used to train AI models must also be taken into careful consideration. The accuracy and relevance of the data used directly affect how the model reacts (Ross, 2022). No algorithm can outperform its training material, therefore if the input data is biased, incorrect, or irrelevant, the AI's output is highly likely to be flawed (Avadhuta, 2020; Lim, 2022). Even when flawed data is acknowledged, it's complicated to establish a "ground truth" (Ross, 2022). The methods that are used to collect data can have severe repercussions on the data itself, which can alter the results. Furthermore, knowing if data is raw or pre-processed and the type of processing that has been conducted is important to consider (Ross, 2022).

If AI is trained on personal data, even more ethical considerations are required to keep this data secure, prevent unauthorized access, and ensure that all privacy norms are followed (Akhtar et al., 2024). This becomes especially important in sensitive domains, such as healthcare, where the use of Electronic Health Records (EHR) calls for certain security protocols to be followed, such as encryption and granular access controls. Questions arise as to who should be the one to manage the data used to train AI; should a public authority control the data, or can it be trusted in the hands of private corporations (Avadhuta, 2020)? In the era of big data, data quickly becomes obsolete, which means that AI models not only need lots of data, but they must be able to parse through their data, remove old data, and replace it with up-to-date information (Lin et al., 2021; Thomasian et al., 2021). On the opposite side of the security spectrum, increases in transparency in data usage, allowing users to see how their data is being used, protecting sensitive data, and building trust is essential for AI developers (Rane et al., 2024; Robert et al., 2020).

3. Unmanned Aerial Vehicle Applications

The field of unmanned aerial vehicles (UAVs), like AI, is a newly emerging technology. UAVs offer significant potential in the field of disaster recovery and healthcare, allowing delivery of supplies to underserved populations and through dangerous conditions. However, the implementation of such technology must include the careful consideration of community concerns, financial sustainability, and ethical implications. Safety, environmental impacts, accessibility, and affordability are all issues that must be considered to prevent the exacerbation of existing inequalities. Resilience is essential for UAV development, especially with multi-UAV systems, or swarms, so that they can work together to solve complex problems. A systems approach that focuses on both equity and resilience is necessary for the responsible and beneficial adoption and implementation of UAV technology (Zhang et al., 2024).

3.1.1 Healthcare Equity with UAVs

UAVs have the potential to improve the accessibility of medical goods and services to underserved areas. In the Democratic Republic of the Congo, where access to medicine is limited, medical delivery drones are emerging as an effective method to deliver supplies where road infrastructure is poorly maintained (O'Sullivan & Leaman, 2022). UAVs were also used during the COVID-19 pandemic, delivering supplies and biological samples to remote locations without requiring direct human interaction (Kemarau et al., 2024). UAVs can reduce the cost of transporting goods to the harder-to-reach areas of the globe. However, with these advances comes a cost; namely privacy and the potential for spying, as well as safety from drones crashing. These concerns can be mitigated with community education, as well as sensitivity training for those piloting the drones (O'Sullivan & Leaman, 2022). The effective scaling of medical delivery drones wouldn't be possible without local government funding or an outside sponsor providing financial aid. Additionally, the lack of technical literacy in developing countries suggests that UAVs would have to be introduced gradually alongside more traditional medical delivery methods to allow for greater communal understanding before moving to fully airborne transport (Gunaratne et al., 2022). It's also important to ensure an equitable practice to determine if supplies are reaching those most in need of them (Eichleay et al., 2019).

3.1.2 Social Equity with UAVs

The integration of UAVs into an urban setting presents a plethora of opportunities as well as risks. Despite UAVs allowing for efficient and fast delivery services, this comes at the expense of underserved communities paying the price of noise and safety risks, further deepening social inequities (Zhang et al., 2024). Both academic research as well as social media analysis reveal there are many aspects of social equity within the realm of urban UAV logistics, some of which are safety, environmental concerns, inclusivity, operations, accessibility, and affordability. Academic literature tends to focus on safety–primarily from collisions–privacy, and noise pollution, while social media suggests that the public is more concerned about accessibility and affordability, with concerns about the loss of jobs taking the forefront (Zhang et al., 2024). Further community education endeavors can help to increase the acceptance of UAV technology into daily life, such as drones performing high-up tasks that are dangerous for humans to perform (Grote et al., 2021).

Ensuring socially equitable practices requires a fair distribution of resources to those who need them most. The resources being distributed not only include the delivery of supplies (covered in the previous section) but also the airspace, allocation of the negative aspects like pollution, and accessibility to services (Zhang et al., 2024). The concept of vertical equity–providing additional resources to vulnerable groups–plays an important role as well. A significant question that must be addressed is regarding the affordability of this technology: will this technology be affordable to low-income communities, or will it always be treated as a luxury reserved for those who can afford it? A profit-motivated UAV model would neglect marginalized communities, which would further disenfranchise them (Zhang et al., 2024). Turning things around, UAVs also raise ethical concerns about surveillance and privacy, as well as direct political and social control, since they can easily be used as tools for reinforcing oppressive power imbalances (Kemarau et al., 2024). To ensure the ethical uses of UAVs, some stakeholders are suggesting regulations that promote privacy and security, and how UAVs will be enforced (Eichleay et al., 2019). The review by Yaqot and Menezes describes some of these applications as "good, bad, and ugly". The "good" side of UAVs explains how they can be used to deliver medical supplies, the "bad" side includes UAVs used for surveillance and to invade privacy, and the "ugly" side of UAVs talks about the types of pollution they could cause, such as filling the airspace and the noise they produce (Yaqot & Menezes, 2021).

3.1.3 Resilience in UAVs

The ability to withstand disruptions, maintain functionality, and remain secure is essential for resilient UAVs, especially multi-UAV systems. Especially within an urban setting, there are various interferences that the system can encounter, from birds, powerlines, and manmade obstructions like an unattended balloon. Resilience for a multi-UAV system would involve real-time evaluation of conditions, understanding the type and severity of disruption, and the ability to make safe decisions autonomously (Ordoukhanian & Madni, 2019). Having a fleet of drones with different capabilities can also increase the resilience of a drone swarm; however, current research is often more concerned with the resilience of individual drones as opposed to a swarm displaying resilient behavior–seeing themselves as a cohesive unit instead of multiple individual drones (Phadke & Medrano, 2022). To fully realize the potential of a drone swarm, the capabilities of a single drone can't simply be amplified to an entire swarm; it would require a need for more comprehensive and resilient cooperation. For example, when searching a disaster site, one drone can search an area with multiple sensors and cameras to check visual feeds, analyze heat signatures, and even listen for people in distress. A swarm of drones could specialize in their specific type of sensing, allowing "listening" drones to fly closer to the ground, and the other two drone variations take higher altitudes and cover more ground at once, allowing for a more efficient rescue endeavor (Phadke & Medrano, 2022). Drones can also be used to predict these kinds of disasters. In the case of node failure within a wireless sensor network, UAVs have been demonstrated to act as a "data mule" to deliver information even when traditional communication methods go dark, allowing for more resilient emergency communication networks (Ueyama et al., 2014).

4. Gaps in the Literature and Future Considerations

While the sources cover multiple domains and topics within equity and resilience, some gaps in the literature exist and should be mentioned. The UN has a framework for promoting equity, called the Sustainable Development Goals (SDG), that details how countries should strive to uplift marginalized groups. SDG 5 in particular aims to "achieve gender equality and empower all women and girls" (Patón-Romero et al., 2022). Currently, no studies address this goal, indicating the lack of development of AI solutions for gender equality. Intellectual property scholarship also does not comprehensively explain the role of AI in IP law, nor are any comprehensive solutions being proposed (Lim, 2022). In healthcare, it's understood that AI bias exists concerning race and sex, but other factors that experience bias, such as gender expression, age, lifestyle, ability, and employment status, require further investigation (Thomasian et al., 2021). A systemic equitable AI for healthcare has yet to exist, despite that methods of mitigating bias exist and are already available (Gurevich et al., 2023). Similarly, despite healthcare AI being designed and passing proof-of-concept testing, there is limited evidence that there's any ecological validity to the models, or that they'd be applicable in real-world scenarios.

In education, the long-term effects of AI integration require further investigation. While AI has the potential to increase the accessibility of education, how this will realistically play out among marginalized and differently abled groups is yet to be seen (Roshanaei et al., 2023). Additionally, we don't yet have a great understanding of how we can blend traditional teaching methods with AI, which will require more research into providing an inclusive, personalized, and adaptive education. The DEI principles referred to within the papers are a Western-dominated concept, and when AI is used in different cultures, there may be a misalignment of traditional values and a misunderstanding of cross-cultural adaptation (Cachat-Rosset & Klarsfeld, 2023). It's also worth noting that there is a notable gap between DEI practices proposed for the ethical use of AI and how AI is developed on a day-to-day basis (Cachat-Rosset & Klarsfeld, 2023).

Focusing back to the AI models themselves, the articles mention often how to avoid biased AI models and ensure that the data being fed to these AI models is fair; what they fail to mention is how an AI system should handle instances of unfairness if they occur, which is a further issue in resilience (Robert et al., 2020). In the context of data collection, a more in-depth analysis is required for the extraction of data when identity and class are relevant to said data (Sadowski, 2019). While disaster resilience in infrastructure is mentioned in some of the articles (Rane et al., 2024), there are few mentions of how AI can help with societal disasters. The articles don't touch on the strategies of how AI can be used to address oppressive power dynamics and unequal access to resources, topics that are essential for societal resilience (Buzzanell, 2023). Current applications for natural disasters already focus on predictive analysis and optimizing resource distribution but won't focus on the root causes of problems and resource disparity.

4.1.1 Future Considerations: The Human Role in AI

Looking towards what can be done differently in future AI endeavors, the most important concept to understand is that while AI appears to act with agency, any technology requires a human touch for human application. To ensure the safe usage of AI tools, there needs to be a clear oversight structure from the development to the deployment of AI models (Ross, 2022). Regulation is required to ensure that bias isn't unintentionally being built into models, as well as to address the barriers to equity throughout the AI lifecycle (Thomasian et al., 2021). Referring to "digital redlining", humans should always be the ones responsible for the final decisions being made. Even as AI becomes more advanced and more autonomous, it's important to leave consequential decisions in the hands of humans, as the human touch of empathy can make a significant impact on the ethical considerations of a problem, as opposed to leaving it all up to algorithmic interpretation of data (Dankwa-Mullan et al., 2021). This also ensures that a particular person or group can be legally or financially held accountable and responsible for the actions of an AI. It's also up to humans to be the ones to identify, prevent, and mitigate bias and discrimination before it becomes part of an AI system (Cachat-Rosset & Klarsfeld, 2023; Lin et al., 2021; Roshanaei et al., 2023). This includes ensuring fair data collection, making explainable and transparent models, hiring diverse teams to monitor the models, and actively searching for and eliminating biases (Hendricks-Sturrup et al., 2023). Creating a system of equity audits can help to hold companies accountable for equitable AI practices and improve the overall accuracy of the models (Lim, 2022).

All ethical guidelines must be provided by humans such that the development of AI tools will uphold the standards of fairness, transparency, and accountability (Akhtar et al., 2024). AI developers are the ones who would be held responsible for their models upholding standards of transparency and explainability (Buzzanell, 2023). Increasing transparency will make understanding AI-generated outcomes simpler, improves error detection, and allows AI users to learn from the mistakes of the models (Ross, 2022). This also encourages greater critical thinking when using AI tools, making users less likely to trust an output at face value. Many professions that use AI, such as within the healthcare industry, must carefully consider their human-computer interactions to prevent harm to their patients. (Ross, 2022). Regular training is required for AI users to keep up their skills, lest they rely too heavily on an automated task; this process is referred to as "deskilling" and shows how important it is for AI users to depend on their intuition, and not to blindly trust outputs. Even an optimal level of trust in AI requires a certain amount of human skepticism. Much like the human factors engineering process, developers of specific AI models need to spend time with the users to understand their needs from the system, and what day-to-day operations look like (Roshanaei et al., 2023). By engaging with professionals in their fields, AI and ML models will better address sociohumanitarian issues and better handle the complexities of equity (Hendricks-Sturrup et al., 2023).

5. Conclusion

The documents presented provide a multifaceted overview of the impact and implications of AI across multiple domains. With a focus on equity and resilience, the documents also touched on DEI, fairness, ethical considerations, and broader societal transformations that are ongoing as well as the future of ethical AI development and usage. Some of the key topics include the domains in which AI is currently being used, resilience with AI, equity with AI, how equity and resilience apply to UAVs, gaps in the current literature, and some considerations for the future. AI is a powerful and expanding technology with the potential to address some of humanity's greatest challenges, such as building resilience and promoting equity. However, in its current state, the development and deployment of AI raises significant concerns with fairness, ethics, equity, and the greater socioeconomic impacts of such a technology. Emphasis should be placed on the need for careful consideration of biased, discriminatory, and unethical data to prevent AI from learning from the mistakes of history. The complex interactions between AI, data, and societal structures require an interdisciplinary effort of diverse groups of developers and auditors to ensure that this powerful tool is not used to further divide our society. When using AI as a tool–much like a kind of superpower–great power comes with great responsibility.

Reference

Abbas, A. (2022). AI & Racial Equity: Understanding Sentiment Analysis Artificial Intelligence, Data Security, and Systemic Theory in Criminal Justice Systems. *arXiv preprint arXiv:2201.00855*.

Akhtar, M. A. K., Kumar, M., & Nayyar, A. (2024). Privacy and Security Considerations in Explainable AI. In *Towards Ethical and Socially Responsible Explainable AI: Challenges and Opportunities* (pp. 193-226). Springer.

Avadhuta, A. S. (2020). The Challenge of Resilience in an Age of Artificial Intelligence. *AI and Robotics in Disaster Studies*, 219-233.

Badarevski, B. (2023). Gender equality and artificial intelligence. *Annual of the Faculty of Philosophy in Skopje*, *76*(1), 805-815.

Buzzanell, P. M. (2023). Risk, resilience, and ethical considerations in artificial intelligence. *Emerging Media*, *1*(1), 30-39.

Cachat-Rosset, G., & Klarsfeld, A. (2023). Diversity, equity, and inclusion in artificial intelligence: an evaluation of guidelines. *Applied Artificial Intelligence*, *37*(1), 2176618.

Carvajal, D., Franco, C., & Isaksson, S. (2024). Will Artificial Intelligence get in the way of achieving gender equality?

Cavazos, S. (2025). The Impact of Artificial Intelligence on Lending: A New Form of Redlining? *Texas A&M Journal of Property Law*, *11*(2), 331-354.

Dankwa-Mullan, I., Scheufele, E. L., Matheny, M. E., Quintana, Y., Chapman, W. W., Jackson, G., & South, B. R. (2021). A proposed framework on integrating health equity and racial justice into the artificial intelligence development lifecycle. *Journal of Health Care for the Poor and Underserved*, *32*(2), 300-317.

Eichleay, M., Evens, E., Stankevitz, K., & Parker, C. (2019). Using the unmanned aerial vehicle delivery decision tool to consider transporting medical supplies via drone. *Global Health: Science and Practice*, *7*(4), 500-506.

Frigerio, I., & Rashidian, N. (2023). Artificial intelligence for equity. *Artificial Intelligence Surgery Editorial Board Study Group on Ethics*, *3*, 160-162.

Grote, M., Pilko, A., Scanlan, J., Cherrett, T., Dickinson, J., Smith, A., Oakey, A., & Marsden, G. (2021). Pathways to unsegregated sharing of airspace: views of the uncrewed aerial vehicle (UAV) industry. *Drones*, *5*(4), 150.

Gunaratne, K., Thibbotuwawa, A., Vasegaard, A. E., Nielsen, P., & Perera, H. N. (2022). Unmanned aerial vehicle adaptation to facilitate healthcare supply chains in low-income countries. *Drones*, *6*(11), 321.

Gurevich, E., El Hassan, B., & El Morr, C. (2023). Equity within AI systems: What can health leaders expect? Healthcare management forum,

Hendricks-Sturrup, R., Simmons, M., Anders, S., Aneni, K., Clayton, E. W., Coco, J., Collins, B., Heitman, E., Hussain, S., & Joshi, K. (2023). Developing ethics and equity principles, terms, and engagement tools to advance health equity and researcher diversity in AI and machine learning: modified Delphi approach. *JMIR AI*, *2*(1), e52888.

Huq, A. Z. (2019). Racial equity in algorithmic criminal justice. *Duke Law Journal*, *68*(6), 1043-1134.

Johnson-Mann, C. N., Loftus, T. J., & Bihorac, A. (2021). Equity and artificial intelligence in surgical care. *JAMA surgery*, *156*(6), 509-510. https://pmc.ncbi.nlm.nih.gov/articles/PMC8273554/pdf/nihms-1721418.pdf

Kemarau, R. A., Sakawi, Z., Suab, S. A., Eboy, O. V., Wan Mohd Jaafar, W. S., Abdul Maulud, K. N., & Md Nor, N. N. F. (2024). Global perspectives on unmanned aerial vehicles technology in social sciences: applications, innovations, and future research directions. *Geocarto International*, *39*(1), 2413547.

Lim, D. (2022). AI, Equity, and the IP Gap. *SMU L. Rev.*, *75*, 815.

Lin, Y.-T., Hung, T.-W., & Huang, L. T.-L. (2021). Engineering equity: How AI can help reduce the harm of implicit bias. *Philosophy & Technology*, *34*(Suppl 1), 65-90.

O'Sullivan, B., & Leaman, M. (2022). Medical delivery drones as a tool to improve health equity in Sub-Saharan Africa. *Global Health: Annual Review*, *1*(7).

Ordoukhanian, E., & Madni, A. M. (2019). Model-based approach to engineering resilience in multi-UAV systems. *Systems*, *7*(1), 11.

Patón-Romero, J. D., Vinuesa, R., Jaccheri, L., & Baldassarre, M. T. (2022). State of gender equality in and by artificial intelligence. *IADIS International Journal on Computer Science and Information Systems*, *17*(2), 31-48.

Phadke, A., & Medrano, F. A. (2022). Towards resilient UAV swarms—A breakdown of resiliency requirements in UAV swarms. *Drones*, *6*(11), 340.

Preece, J., Rogers, Y., & Sharp, H. (2023). Interaction Design: beyond human-computer interaction. In (6th ed.). Wiley.

Rane, N., Choudhary, S., & Rane, J. (2024). Artificial intelligence for enhancing resilience. *Journal of Applied Artificial Intelligence*, *5*(2), 1-33.

Renski, H., Smith-Doerr, L., Wilkerson, T., Roberts, S. C., Zilberstein, S., & Branch, E. H. (2020). Racial Equity and the Future of Work. *Technology| Architecture+ Design*, *4*(1), 17-22.

Robert, L. P., Pierce, C., Marquis, L., Kim, S., & Alahmad, R. (2020). Designing fair AI for managing employees in organizations: a review, critique, and design agenda. *Human–Computer Interaction*, *35*(5-6), 545-575.

Roshanaei, M., Olivares, H., & Lopez, R. R. (2023). Harnessing AI to foster equity in education: Opportunities, challenges, and emerging strategies. *Journal of Intelligent Learning Systems and Applications*, *15*(4), 123-143.

Ross, P. (2022). Racial Equity Implications of Artificial Intelligence in Health Care. *Food & Drug LJ*, *77*, 265.

Sadowski, J. (2019). When data is capital: Datafication, accumulation, and extraction. *Big data & society*, *6*(1), 2053951718820549.

Thomasian, N. M., Eickhoff, C., & Adashi, E. Y. (2021). Advancing health equity with artificial intelligence. *Journal of public health policy*, *42*(4), 602. https://pmc.ncbi.nlm.nih.gov/articles/PMC8607970/pdf/41271_2021_Article_319.pdf

Ueyama, J., Freitas, H., Faiçal, B. S., PR Filho, G., Fini, P., Pessin, G., Gomes, P. H., & Villas, L. A. (2014). Exploiting the use of unmanned aerial vehicles to provide resilience in wireless sensor networks. *IEEE Communications Magazine*, *52*(12), 81-87.

Woods, D. D. (2015). Four concepts for resilience and the implications for the future of resilience engineering. *Reliability Engineering & System Safety*, *141*, 5-9.

Yaqot, M., & Menezes, B. (2021). The good, the bad, and the ugly: review on the social impacts of unmanned aerial vehicles (UAVs). International Conference of Reliable Information and Communication Technology,

Zhang, D., Yang, P. P.-J., & Tsou, J.-Y. (2024). Advancing Social Equity in Urban UAV Logistics: Insights from the Academic Literature and Social Media. *Drones*, *8*(11), 688.